

In [44]: `import pandas as pd`

In [45]: `data = pd.read_csv('PQ Data.csv')`

In [46]: `data`

Out[46]:

	Date	Customer Age	Customer Gender	Country	State	Product Category	Sub Category	Quantity
0	2/19/2016	29	F	United States	Washington	Accessories	Tires and Tubes	
1	2/20/2016	29	F	United States	Washington	Clothing	Gloves	
2	2/27/2016	29	F	United States	Washington	Accessories	Tires and Tubes	
3	3/12/2016	29	F	United States	Washington	Accessories	Tires and Tubes	
4	3/12/2016	29	F	United States	Washington	Accessories	Tires and Tubes	
...	...	...	...	...	...	...	...	...
34860	3/22/2015	38	M	France	Charente-Maritime	Bikes	Mountain Bikes	
34861	2/7/2016	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	
34862	3/13/2015	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	
34863	4/5/2015	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	
34864	8/30/2015	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	

34865 rows × 10 columns



In [47]: `# head()`
`# tail()`
`# sample()`
`# info()`

In [48]: `data.head(60)`

Out[48]:

	Date	Customer Age	Customer Gender	Country	State	Product Category	Sub Category	Quantity
0	2/19/2016	29	F	United States	Washington	Accessories	Tires and Tubes	1
1	2/20/2016	29	F	United States	Washington	Clothing	Gloves	1
2	2/27/2016	29	F	United States	Washington	Accessories	Tires and Tubes	1
3	3/12/2016	29	F	United States	Washington	Accessories	Tires and Tubes	1
4	3/12/2016	29	F	United States	Washington	Accessories	Tires and Tubes	1
5	4/8/2016	29	F	United States	Washington	Accessories	Tires and Tubes	1
6	4/17/2016	29	F	United States	Washington	Accessories	Tires and Tubes	1
7	4/17/2016	29	F	United States	Washington	Accessories	Tires and Tubes	1
8	6/22/2016	29	F	United States	Washington	Accessories	Tires and Tubes	1
9	6/22/2016	29	F	United States	Washington	Accessories	Tires and Tubes	1
10	6/24/2016	29	F	United States	Washington	Accessories	Helmets	1
11	7/1/2016	29	F	United States	Washington	Accessories	Tires and Tubes	1
12	7/22/2016	29	F	United States	Washington	Accessories	Tires and Tubes	1
13	7/22/2016	29	F	United States	Washington	Accessories	Tires and Tubes	1
14	8/18/2015	29	F	United States	Washington	Accessories	Helmets	1
15	8/30/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
16	9/17/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
17	9/17/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
18	9/17/2015	29	F	United States	Washington	Accessories	Helmets	1

	Date	Customer Age	Customer Gender	Country	State	Product Category	Sub Category	Quantity
19	9/24/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
20	9/24/2015	29	F	United States	Washington	Accessories	Helmets	1
21	10/7/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
22	10/7/2015	29	F	United States	Washington	Accessories	Helmets	1
23	10/7/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
24	10/7/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
25	10/13/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
26	10/20/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
27	10/20/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
28	10/20/2015	29	F	United States	Washington	Accessories	Helmets	1
29	11/1/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
30	11/1/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
31	11/1/2015	29	F	United States	Washington	Accessories	Helmets	1
32	11/15/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
33	11/15/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
34	11/15/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
35	12/15/2015	29	F	United States	Washington	Accessories	Helmets	1
36	12/16/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
37	12/16/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1

	Date	Customer Age	Customer Gender	Country	State	Product Category	Sub Category	Quantity
38	12/16/2015	29	F	United States	Washington	Accessories	Helmets	1
39	12/20/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
40	12/20/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
41	12/20/2015	29	F	United States	Washington	Accessories	Helmets	1
42	12/25/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
43	12/25/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
44	12/25/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
45	12/29/2015	29	F	United States	Washington	Accessories	Tires and Tubes	1
46	12/29/2015	29	F	United States	Washington	Accessories	Helmets	1
47	7/22/2015	18	F	United States	California	Accessories	Bike Stands	1
48	10/10/2015	18	F	United States	California	Accessories	Bike Stands	1
49	6/11/2016	18	M	United States	California	Bikes	Mountain Bikes	1
50	8/13/2015	18	M	United States	California	Bikes	Mountain Bikes	1
51	9/10/2015	18	M	United States	California	Bikes	Mountain Bikes	1
52	10/20/2015	18	M	United States	California	Bikes	Mountain Bikes	1
53	12/21/2015	18	M	United States	California	Bikes	Mountain Bikes	1
54	7/27/2015	19	F	United States	Oregon	Accessories	Hydration Packs	1
55	2/7/2016	19	M	United States	Washington	Bikes	Mountain Bikes	1
56	7/24/2015	19	M	United States	Washington	Bikes	Mountain Bikes	1

	Date	Customer Age	Customer Gender	Country	State	Product Category	Sub Category	Quantity
57	3/20/2016	19	M	United States	California	Clothing	Jerseys	1
58	3/21/2016	19	M	United States	California	Clothing	Jerseys	1
59	5/6/2016	19	M	United States	California	Clothing	Jerseys	1

In [49]: `data.tail()`

Out[49]:

	Date	Customer Age	Customer Gender	Country	State	Product Category	Sub Category	Quantity
34860	3/22/2015	38	M	France	Charente-Maritime	Bikes	Mountain Bikes	1
34861	2/7/2016	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	2
34862	3/13/2015	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	1
34863	4/5/2015	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	3
34864	8/30/2015	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	1



In [50]: `data.sample()`

Out[50]:

	Date	Customer Age	Customer Gender	Country	State	Product Category	Sub Category	Quantity
14352	2/5/2016	43	M	United States	California	Clothing	Caps	1



In [51]: `data.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 34865 entries, 0 to 34864
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Date                  34865 non-null object
1   Customer Age          34865 non-null int64
2   Customer Gender       34865 non-null object
3   Country               34865 non-null object
4   State                 34865 non-null object
5   Product Category      34865 non-null object
6   Sub Category          34865 non-null object
7   Quantity              34865 non-null int64
8   Unit Cost             34865 non-null object
9   Unit Price            34865 non-null object
dtypes: int64(2), object(8)
memory usage: 2.7+ MB
```

```
In [52]: data.Date = pd.to_datetime(data.Date)
```

```
In [53]: data.Date.dt.year
```

```
Out[53]: 0      2016
1      2016
2      2016
3      2016
4      2016
...
34860   2015
34861   2016
34862   2015
34863   2015
34864   2015
Name: Date, Length: 34865, dtype: int32
```

```
In [54]: data["year"] = data.Date.dt.year
```

```
In [55]: data.sample()
```

```
Out[55]:
```

	Date	Customer Age	Customer Gender	Country	State	Product Category	Sub Category	Quantity	Unit Cost
10227	2015-12-29	37	F	United Kingdom	England	Clothing	Vests	3	\$38

```
In [56]: data["Unit Cost"] = data["Unit Cost"].str.replace("$" , "").str.replace(",","")
```

```
In [57]: data["Unit Cost"] = data["Unit Cost"].astype("int")
```

```
In [58]: data["Unit Price"] = data["Unit Price"].str.replace("$" , "").str.replace(",","")
```

```
In [59]: data["Unit Price"] = data["Unit Price"].astype("int")
```

```
In [60]: data["total_sales"] = data.Quantity * data["Unit Price"]
```

```
In [61]: data["total_cost"] = data.Quantity * data["Unit Cost"]
```

```
In [62]: data["profit"] = data.total_sales - data.total_cost
```

```
In [63]: data
```

Out[63]:

	Date	Customer Age	Customer Gender	Country	State	Product Category	Sub Category	Quantity
0	2016-02-19	29	F	United States	Washington	Accessories	Tires and Tubes	1
1	2016-02-20	29	F	United States	Washington	Clothing	Gloves	2
2	2016-02-27	29	F	United States	Washington	Accessories	Tires and Tubes	3
3	2016-03-12	29	F	United States	Washington	Accessories	Tires and Tubes	3
4	2016-03-12	29	F	United States	Washington	Accessories	Tires and Tubes	3
...	...	...	...	...	...	...	...	...
34860	2015-03-22	38	M	France	Charente-Maritime	Bikes	Mountain Bikes	1
34861	2016-02-07	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	2
34862	2015-03-13	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	1
34863	2015-04-05	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	3
34864	2015-08-30	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	1

34865 rows × 14 columns

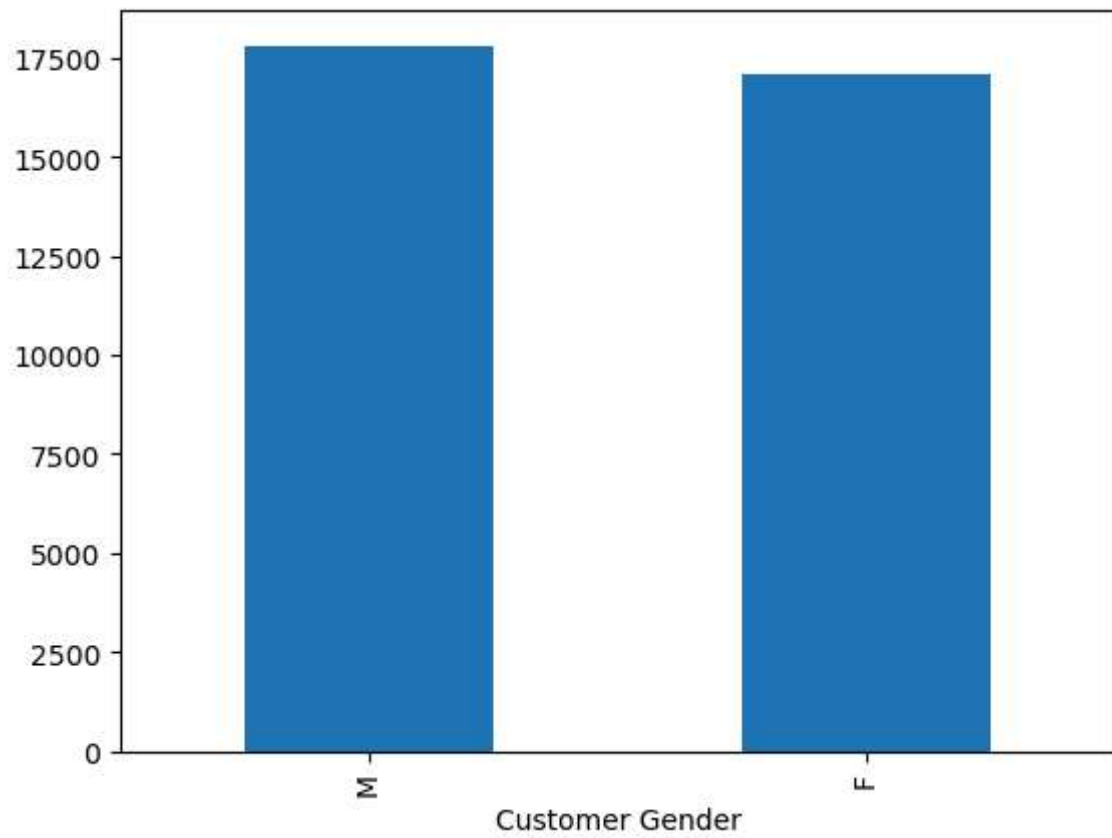
```
In [64]: data.sample()
```

Out[64]:

	Date	Customer Age	Customer Gender	Country	State	Product Category	Sub Category	Quantity	U C
28284	2016-05-19	30	F	United States	California	Accessories	Tires and Tubes	2	

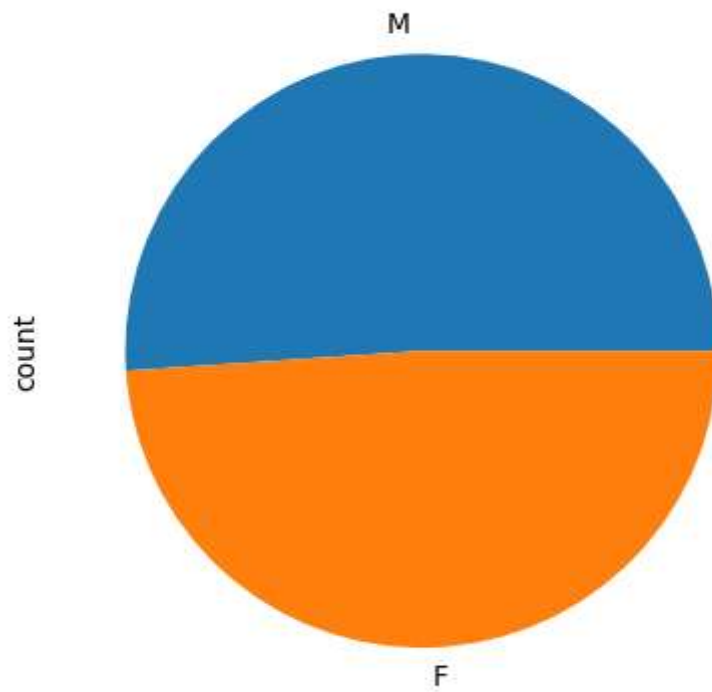
```
In [65]: data["Customer Gender"].value_counts().plot(kind = "bar")
```

```
Out[65]: <Axes: xlabel='Customer Gender'>
```



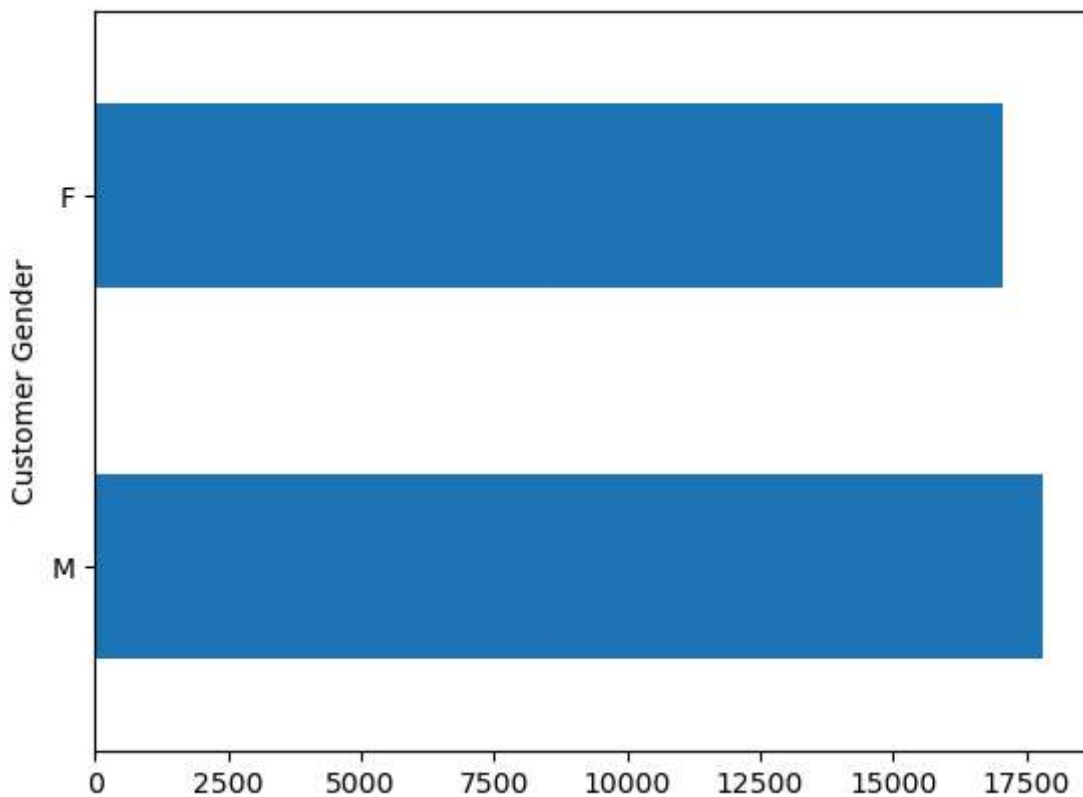
```
In [66]: data["Customer Gender"].value_counts().plot(kind = "pie")
```

```
Out[66]: <Axes: ylabel='count'>
```

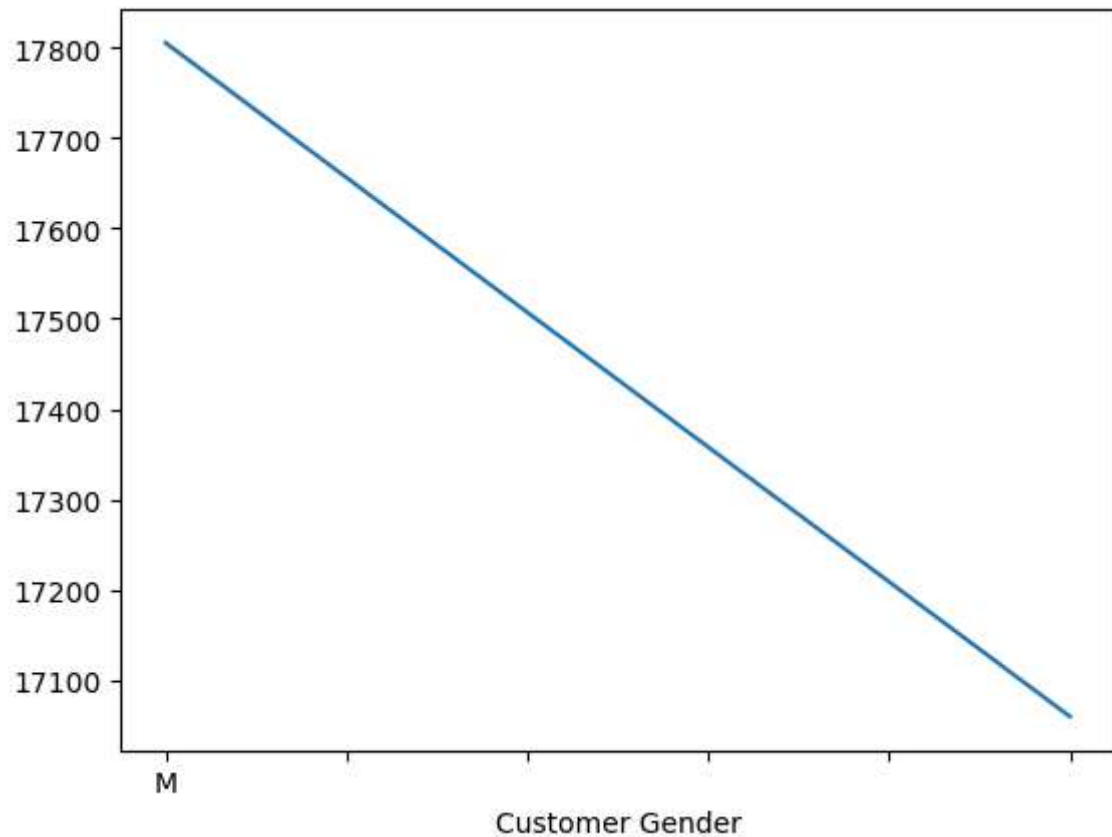
```
In [67]: data["Customer Gender"].value_counts().plot(kind = "barh")
```

```
Out[67]: <Axes: ylabel='Customer Gender'>
```



```
In [68]: data["Customer Gender"].value_counts().plot(kind = "line")
```

```
Out[68]: <Axes: xlabel='Customer Gender'>
```



In [69]: `data.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 34865 entries, 0 to 34864
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   Date                  34865 non-null  datetime64[ns]
 1   Customer Age          34865 non-null  int64  
 2   Customer Gender       34865 non-null  object  
 3   Country                34865 non-null  object  
 4   State                 34865 non-null  object  
 5   Product Category      34865 non-null  object  
 6   Sub Category          34865 non-null  object  
 7   Quantity              34865 non-null  int64  
 8   Unit Cost              34865 non-null  int64  
 9   Unit Price            34865 non-null  int64  
10  year                  34865 non-null  int32  
11  total_sales           34865 non-null  int64  
12  total_cost            34865 non-null  int64  
13  profit                34865 non-null  int64  
dtypes: datetime64[ns](1), int32(1), int64(7), object(5)
memory usage: 3.6+ MB
```

In [70]: `data["Product Category"].value_counts()`

```
Out[70]: Product Category
Accessories    22533
Bikes          7093
Clothing       5239
Name: count, dtype: int64
```

```
In [71]: data.sample()
```

```
Out[71]:
```

	Date	Customer Age	Customer Gender	Country	State	Product Category	Sub Category	Quantity	Unit Cost
10307	2015-08-25	28	F	France	Essonne	Bikes	Road Bikes	1	2443



```
In [72]: data.to_csv("pd_clean.csv")
```

```
In [73]: data.describe()
```

```
Out[73]:
```

	Date	Customer Age	Quantity	Unit Cost	Unit Price	
count	34865	34865.000000	34865.000000	34865.000000	34865.000000	34865.0
mean	2016-01-19 18:41:43.530761472	36.382705	2.002553	349.948860	389.325541	2015.5
min	2015-01-01 00:00:00	17.000000	1.000000	1.000000	1.000000	2015.0
25%	2015-10-26 00:00:00	28.000000	1.000000	45.000000	54.000000	2015.0
50%	2016-01-28 00:00:00	35.000000	2.000000	150.000000	179.000000	2016.0
75%	2016-04-26 00:00:00	44.000000	3.000000	455.000000	521.000000	2016.0
max	2016-07-31 00:00:00	87.000000	3.000000	3240.000000	5082.000000	2016.0
std	NaN	11.113005	0.813965	490.010436	525.311617	0.4



```
In [74]: data.total_sales.describe() # median if is close to mean so there is no outlier
```

```
Out[74]: count    34865.000000
         mean      641.054353
         std       736.651149
         min        2.000000
         25%       102.000000
         50%       319.000000
         75%       903.000000
         max      5082.000000
         Name: total_sales, dtype: float64
```

```
In [75]: data.drop_duplicates(inplace=True)
```

```
In [76]: data.total_sales
```

```
Out[76]: 0         109
         1          58
         2          15
         3         351
         4         126
         ...
        34860      1487
        34861      1972
        34862      1583
        34863      1683
        34864      1568
         Name: total_sales, Length: 34861, dtype: int64
```

```
In [77]: def saled_calssification(sales_value):    # user defined function
         if sales_value <= 320 :
             return "low sales"
         else :
             return "high sales"
```

```
In [79]: data['saled_calssification']=data['total_sales'].apply(saled_calssification)
```

```
In [80]: data
```

Out[80]:

	Date	Customer Age	Customer Gender	Country	State	Product Category	Sub Category	Quantity
0	2016-02-19	29	F	United States	Washington	Accessories	Tires and Tubes	1
1	2016-02-20	29	F	United States	Washington	Clothing	Gloves	2
2	2016-02-27	29	F	United States	Washington	Accessories	Tires and Tubes	3
3	2016-03-12	29	F	United States	Washington	Accessories	Tires and Tubes	3
4	2016-03-12	29	F	United States	Washington	Accessories	Tires and Tubes	3
...	...	...	...	...	...	...	...	...
34860	2015-03-22	38	M	France	Charente-Maritime	Bikes	Mountain Bikes	1
34861	2016-02-07	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	2
34862	2015-03-13	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	1
34863	2015-04-05	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	3
34864	2015-08-30	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	1

34861 rows × 15 columns



```
In [86]: def profit_classification(profit_value): # user defined function m must there is n
        if profit_value < 0:
            return "Negative"
        elif profit_value > 0:
            return "Positive"
        else:
            return "Break Even"
```

```
In [88]: data['profit_classification']=data['profit'].apply(profit_classification)
```

```
In [89]: data
```

Out[89]:

	Date	Customer Age	Customer Gender	Country	State	Product Category	Sub Category	Quantity
0	2016-02-19	29	F	United States	Washington	Accessories	Tires and Tubes	1
1	2016-02-20	29	F	United States	Washington	Clothing	Gloves	2
2	2016-02-27	29	F	United States	Washington	Accessories	Tires and Tubes	3
3	2016-03-12	29	F	United States	Washington	Accessories	Tires and Tubes	3
4	2016-03-12	29	F	United States	Washington	Accessories	Tires and Tubes	3
...	...	...	...	...	...	...	...	...
34860	2015-03-22	38	M	France	Charente-Maritime	Bikes	Mountain Bikes	1
34861	2016-02-07	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	2
34862	2015-03-13	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	1
34863	2015-04-05	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	3
34864	2015-08-30	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	1

34861 rows × 16 columns



```
In [92]: data['profit_classification'] = data['profit'].apply(
         lambda profit_value: "Pos" if profit_value > 0 else ("Neg" if profit_value < 0
         )
```

```
In [93]: data
```

Out[93]:

	Date	Customer Age	Customer Gender	Country	State	Product Category	Sub Category	Quantity
0	2016-02-19	29	F	United States	Washington	Accessories	Tires and Tubes	1
1	2016-02-20	29	F	United States	Washington	Clothing	Gloves	2
2	2016-02-27	29	F	United States	Washington	Accessories	Tires and Tubes	3
3	2016-03-12	29	F	United States	Washington	Accessories	Tires and Tubes	3
4	2016-03-12	29	F	United States	Washington	Accessories	Tires and Tubes	3
...	...	...	...	...	...	...	...	...
34860	2015-03-22	38	M	France	Charente-Maritime	Bikes	Mountain Bikes	1
34861	2016-02-07	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	2
34862	2015-03-13	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	1
34863	2015-04-05	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	3
34864	2015-08-30	38	M	France	Hauts de Seine	Bikes	Mountain Bikes	1

34861 rows × 16 columns



In [96]:

```
data.pivot_table(  
    index="Product Category",  
    values="total_sales",  
    aggfunc="sum"  
)
```

Out[96]:

total_sales	
Product Category	
Accessories	7424334
Bikes	11487481
Clothing	3438372

In [99]:

```
data.pivot_table(  
    index=["Product Category", "Country"],  
    values="profit",
```

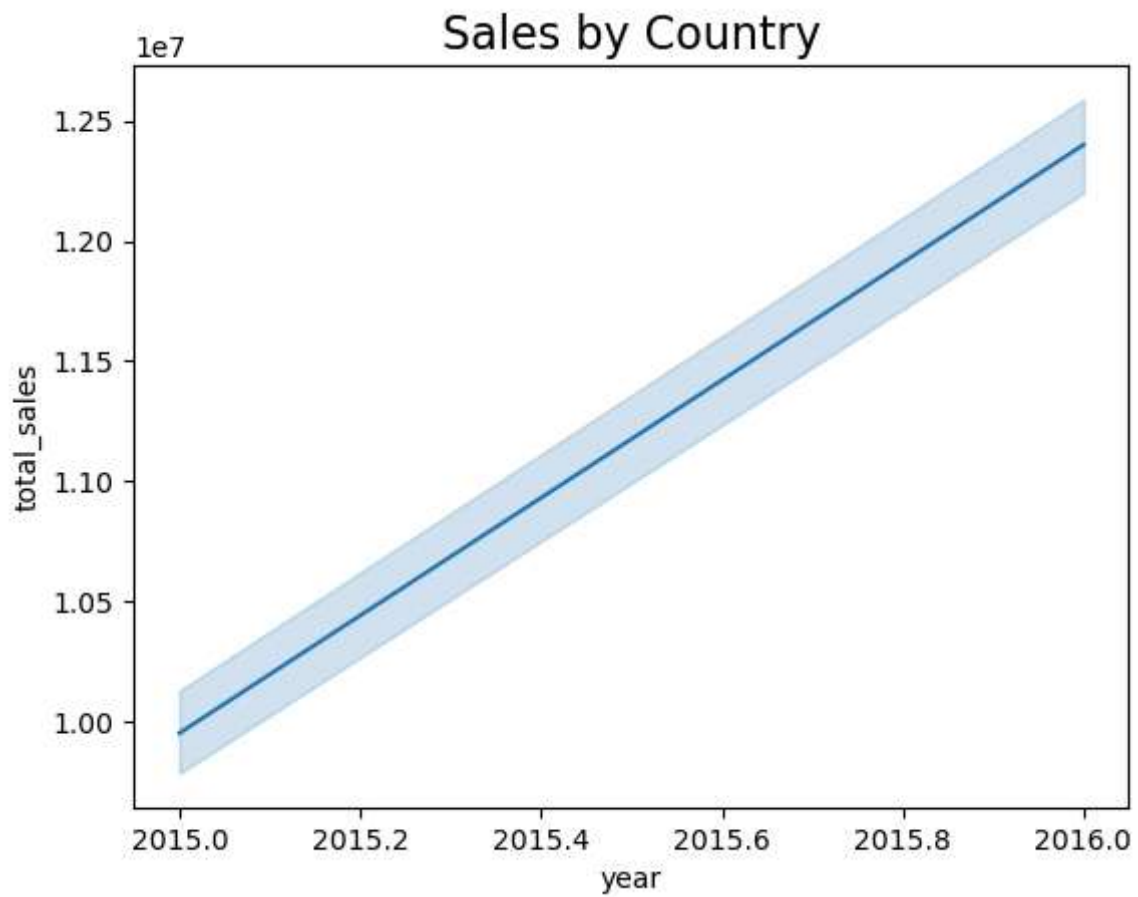
```
aggfunc="mean"
)
```

Out[99]:

		profit
Product Category	Country	
Accessories	France	58.842345
	Germany	124.610694
	United Kingdom	56.964375
	United States	46.464698
Bikes	France	12.860243
	Germany	322.415182
	United Kingdom	10.602538
	United States	-34.363463
Clothing	France	89.016598
	Germany	203.412676
	United Kingdom	92.134328
	United States	87.302999

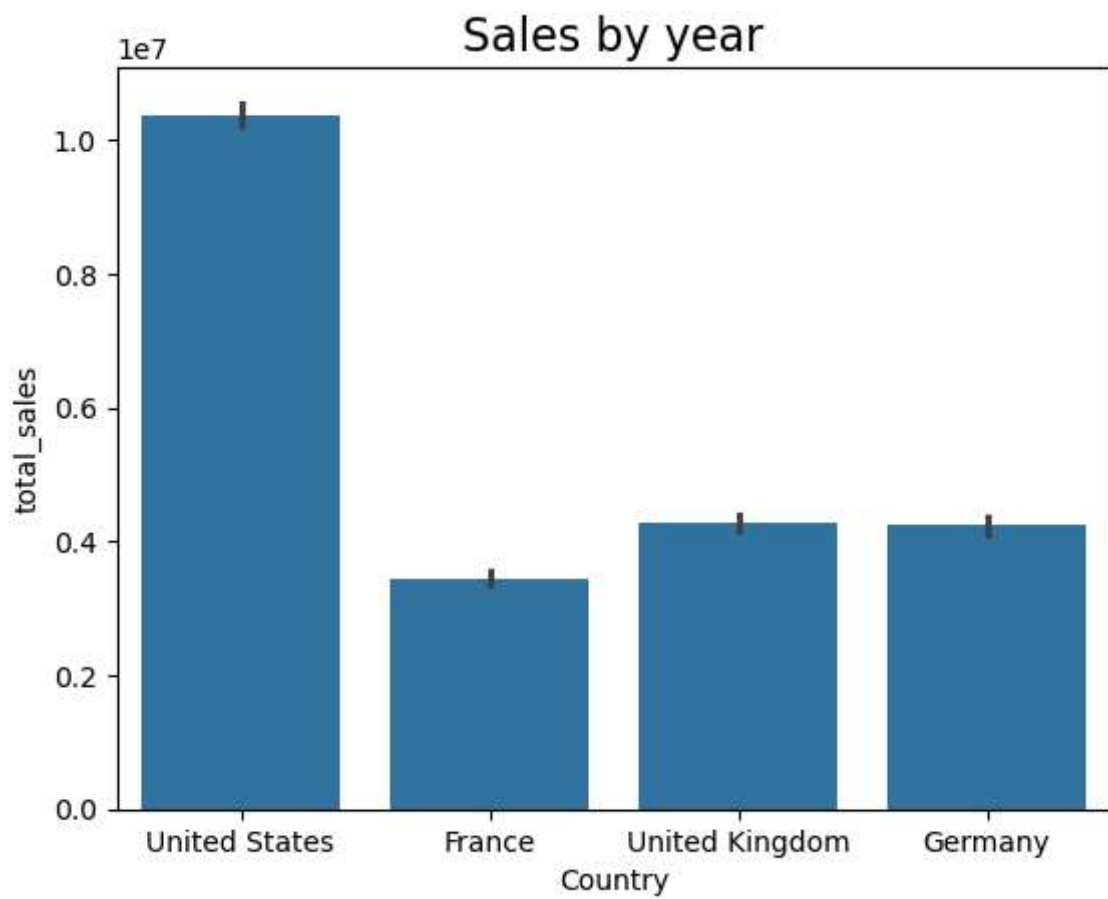
```
In [104... import seaborn as sns # charts
import matplotlib.pyplot as plt # title
```

```
In [107... plt.title("Sales by Country", fontsize=16)
sns.lineplot(data=data, x="year", y="total_sales", estimator="sum")
plt.show() # to delete the the words above the chart
```

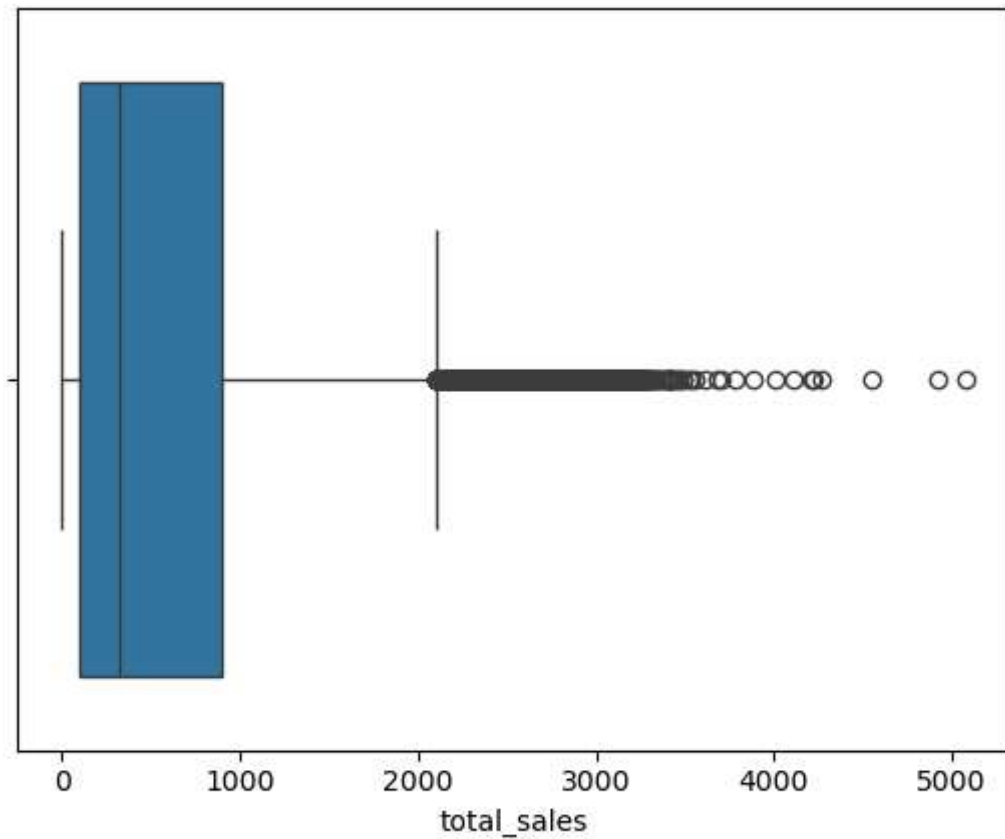
In [108...

```
plt.title("Sales by year", fontsize=16)
sns.barplot(data=data, x="Country", y="total_sales", estimator="sum")
plt.show() # to delete the the words above the chart
```



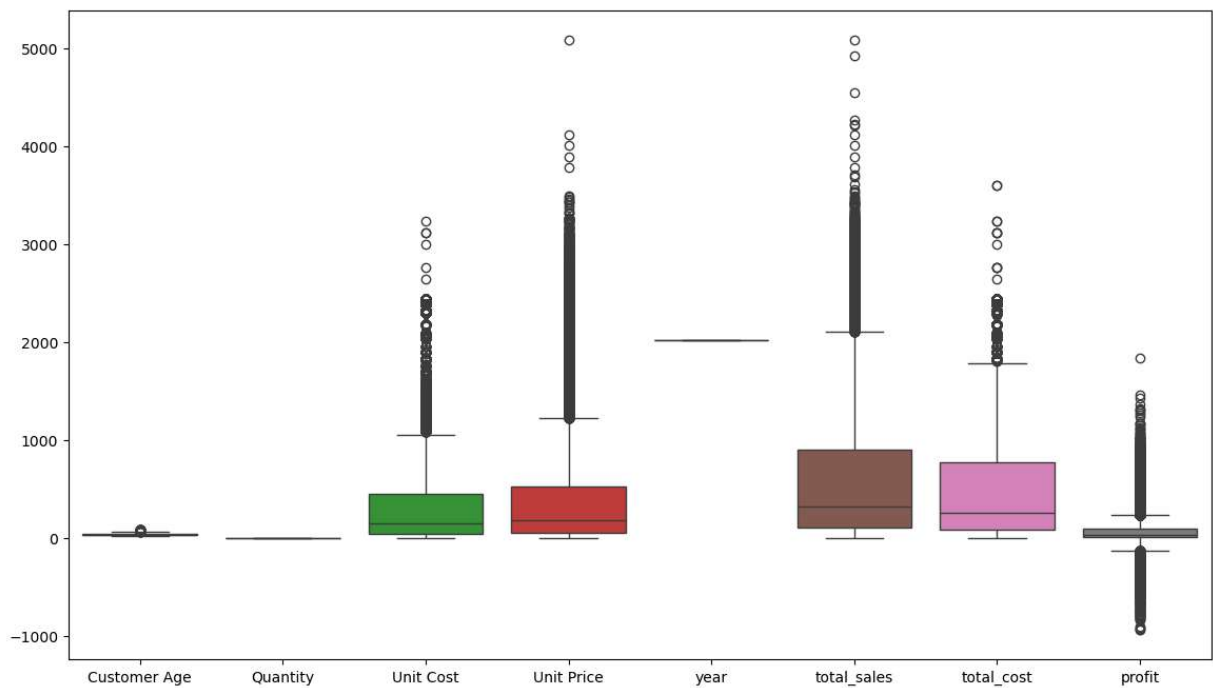
```
In [110...] sns.boxplot(data= data ,x = "total_sales" )
```

```
Out[110...] <Axes: xlabel='total_sales'>
```



In [113... `plt.figure(figsize=(14, 8))` # حدد الحجم الأول
`sns.boxplot(data=data)`

Out[113... <Axes: >



In [118... `data.select_dtypes("object")`

Out[118...

	Customer Gender	Country	State	Product Category	Sub Category	saled_calssification	profit_cl
0	F	United States	Washington	Accessories	Tires and Tubes	low sales	
1	F	United States	Washington	Clothing	Gloves	low sales	
2	F	United States	Washington	Accessories	Tires and Tubes	low sales	
3	F	United States	Washington	Accessories	Tires and Tubes	high sales	
4	F	United States	Washington	Accessories	Tires and Tubes	low sales	
...	...	...	...	...	...	...	
34860	M	France	Charente-Maritime	Bikes	Mountain Bikes	high sales	
34861	M	France	Hauts de Seine	Bikes	Mountain Bikes	high sales	
34862	M	France	Hauts de Seine	Bikes	Mountain Bikes	high sales	
34863	M	France	Hauts de Seine	Bikes	Mountain Bikes	high sales	
34864	M	France	Hauts de Seine	Bikes	Mountain Bikes	high sales	

34861 rows × 7 columns



In [116...

```
data.select_dtypes("number").corr()
```

Out[116...

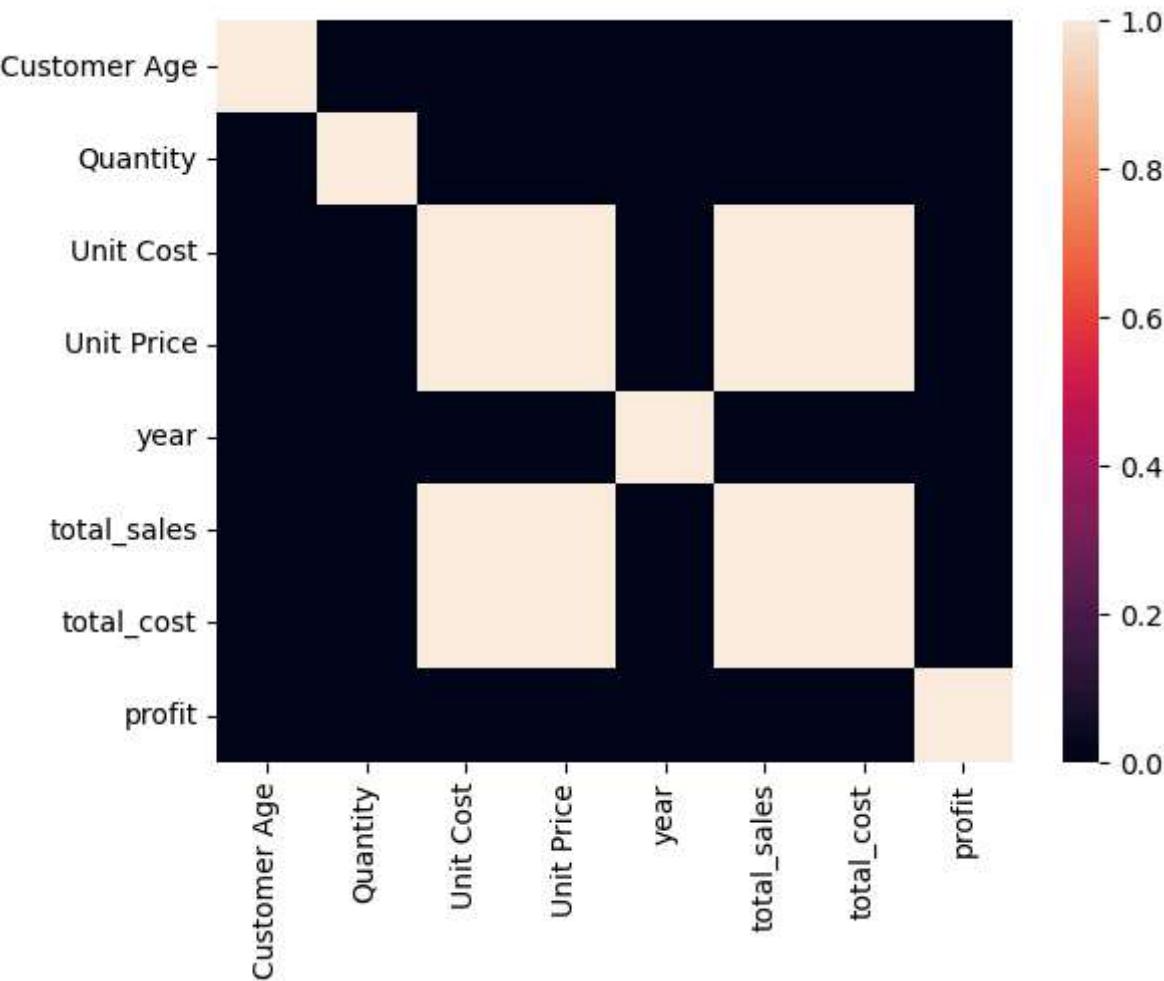
	Customer Age	Quantity	Unit Cost	Unit Price	year	total_sales	total_cost	
Customer Age	1.000000	0.004400	0.022311	0.023359	0.049884	0.024774	0.023582	0.0
Quantity	0.004400	1.000000	-0.312527	-0.324106	0.001030	0.008454	0.008294	0.0
Unit Cost	0.022311	-0.312527	1.000000	0.981030	-0.073332	0.836915	0.854900	0.1
Unit Price	0.023359	-0.324106	0.981030	1.000000	-0.022709	0.851010	0.832974	0.3
year	0.049884	0.001030	-0.073332	-0.022709	1.000000	-0.025448	-0.084665	0.2
total_sales	0.024774	0.008454	0.836915	0.851010	-0.025448	1.000000	0.979116	0.3
total_cost	0.023582	0.008294	0.854900	0.832974	-0.084665	0.979116	1.000000	0.2
profit	0.012865	0.003280	0.171528	0.338459	0.259737	0.396313	0.201381	1.0

In [117...

```
sns.heatmap(data = data.select_dtypes("number").corr() > 0.7)
```

Out[117...

<Axes: >



In []: